

Auditable Day-Ahead Electricity Price Forecasting: Leakage Control, Conditional Failure Maps, and Walk-Forward Evidence from Germany–Luxembourg

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Abstract

Day-ahead electricity-price models are unusually easy to overstate. Prices are spiky and can be negative; twenty-four hourly products clear jointly; forecast errors are strongly dependent within a delivery day; and public archives do not always preserve the information vintage that would have been available before gate closure. This paper presents a deliberately compact, fully executable forecasting study for the Germany–Luxembourg (DE-LU) day-ahead market. The artifact combines 15,576 hourly observations from 2024-08-31 to 2026-06-11 with day-ahead wind and solar forecasts, lagged prices, and calendar variables. Its primary claim excludes weather covariates whose public historical archive does not preserve a fixed D-1 lead time.

Four point forecasts are evaluated on a strict expanding-window experiment covering 365 delivery days and 8,760 hourly prices. Models are refitted every seven days, and all observations from a local delivery day remain in the same train or test block. The weekly naive forecast obtains an MAE of 32.89 EUR/MWh, pooled Ridge obtains 17.11, a 24-model hourly Ridge obtains 15.73, and deterministic LightGBM obtains 13.59 with a seven-day moving-block bootstrap 95% interval of [12.39, 14.86]. The corresponding improvement is 13.6% relative to the strongest linear baseline. Daily absolute-loss differentials remain significant under a seven-lag heteroskedasticity- and autocorrelation-consistent variance estimate with a Harvey–Leybourne–Newbold finite-sample correction. Feature-family ablations show that removing day-ahead renewables raises MAE by 84.9%, whereas adding the ambiguous-vintage weather proxy changes MAE by only -0.81%.

Uncertainty is reported rather than inferred from point accuracy. A strictly prequential, rolling-residual interval achieves 90.07% empirical coverage against a 90% nominal target, with mean width 59.03 EUR/MWh. This marginal result masks a formal hour-conditional failure: 18 of 24 hours differ from nominal coverage under an unadjusted normal test and 16 remain significant after Holm correction. Tail diagnostics show that MAE reaches 21.59 on negative-price hours and 61.12 above 200 EUR/MWh. A separate prompt-proxy experiment is explicitly bounded by no-signal, naive, linear-model, and infeasible perfect-day references. It carries direction beyond weekly persistence, but much of its raw 74.3% hit rate is mechanical. It is not presented as tradable P&L.

The contribution is methodological rather than a claim of market dominance: the repository binds every table and figure to frozen CSV evidence, exposes data-vintage limitations, includes executable tests, and builds an arXiv source archive without private data or credentials.

Keywords: electricity price forecasting; day-ahead market; DE-LU; LightGBM; renewable forecasts; conformal intervals; walk-forward validation; reproducible research.

1 Introduction

European day-ahead electricity prices are the output of a physical system and a market-clearing mechanism. Supply and demand must balance continuously, storage remains limited relative to system scale, renewable production is weather-dependent, and marginal generation costs can change abruptly across hours. A useful forecasting system must therefore do more than minimize a single

error metric. It must specify exactly when each input was available, respect the daily auction structure, compare against credible baselines, and quantify uncertainty under serial dependence.

These requirements are familiar in the electricity-price forecasting literature. Weron [19] emphasizes common data and robust comparative evaluation, while Lago et al. [13] argue that short test samples, weak baselines, and missing significance tests make many model comparisons difficult to interpret. Machine-learning methods can capture nonlinear interactions in price history and fundamentals [10], but their flexibility also creates more opportunities for hidden look-ahead, repeated tuning, and selective reporting.

This paper turns an operational prototype into a reviewable computational artifact. It forecasts all hourly DE-LU prices for delivery day d using information intended to be available before the D-1 coupled auction. The same repository performs ingestion, quality assurance, feature construction, walk-forward estimation, statistical diagnostics, visualization, and paper generation. Frozen data and predictions are committed so that a reviewer can audit the reported values without repeatedly calling an external API.

1.1 Research questions

The study is organized around five questions.

1. Can a compact nonlinear model improve materially on a weekly naive forecast, a pooled regularized linear baseline, and a stronger 24-model hourly Ridge baseline over a complete out-of-sample year?
2. Does the apparent improvement survive delivery-day uncertainty intervals and autocorrelation-robust loss-differential tests?
3. Which feature families account for the improvement, and does a weather-augmented sensitivity materially change the answer?
4. How does error vary across seasons, hours, renewable regimes, negative prices, and high-price events?
5. Can rolling residuals provide useful prequential prediction intervals, and what does a separate prompt-proxy translation reveal without claiming a backtested trading strategy?

1.2 Contributions

The artifact makes six concrete contributions.

1. It publishes a complete one-year, hourly expanding-window experiment with three baselines, including 24 delivery-hour Ridge models, under a shared information set.
2. It treats the local delivery day, not the individual hour, as the primary dependence block for confidence intervals and loss comparisons.
3. It separates a defensible D-1 primary feature set from a weather sensitivity whose archive does not preserve a fixed day-ahead vintage.
4. It reports deterministic feature-family ablations, regime-specific errors, extreme-price detection rates, and a formal map of conditional interval failure.
5. It corrects right-censoring in the prompt-proxy evaluation and applies a fourteen-day moving-block bootstrap to overlapping forward-week outcomes.
6. It distributes source code, tests, frozen evidence, vector figures, metadata, and a self-contained arXiv source package.

No claim is made that LightGBM is the best available DE-LU model, that the sample spans all market regimes, or that the prompt-proxy results are monetizable. The narrower claim is that the declared implementation outperforms its three declared baselines on this declared sample, and that the repository exposes enough evidence to assess that claim.

2 Market setting and related work

2.1 The DE-LU day-ahead task

In the coupled European day-ahead market, bids for delivery day d are submitted on day $d - 1$ and the hourly products clear jointly around midday. The EPEX SPOT market timeline places the European hourly coupled auction at 12:00 CET [3]. For a forecasting process run before that gate closure, the target is the vector

$$\mathbf{y}_d = (y_{d,1}, \dots, y_{d,H_d})^\top,$$

where H_d is 23, 24, or 25 on daylight-saving transition days. The artifact stores timestamps in UTC and derives market behavior from Europe/Berlin local time. This avoids duplicated UTC timestamps while preserving the local calendar actually used by the market.

Prices may be negative when inflexible supply and renewable generation exceed short-run demand, while scarcity or outages can generate positive spikes. The present data contain 5.92% negative-price hours. Consequently, percentage errors such as MAPE are inappropriate: the denominator can be zero or negative, and extreme ratios can dominate the statistic. The primary score is absolute error in EUR/MWh, supplemented by RMSE, bias, tail quantiles, and regime-specific diagnostics.

2.2 Forecasting literature

Electricity-price forecasting spans structural, statistical, and machine-learning approaches [19]. The present design is not a structural dispatch model. It learns a reduced-form map from observable history and day-ahead renewable forecasts to hourly prices. Ridge regression [9] provides a stable linear benchmark after standardization. Gradient boosting constructs an additive nonlinear predictor through stagewise fitting [5]; LightGBM implements efficient histogram-based gradient-boosted trees [11].

The choice of a nonlinear tree model is pragmatic. Price responses to renewable output can be state-dependent, hour effects are not smooth across the entire day, and a fixed linear slope may miss interactions between recent price levels and the day-over-day renewable change. This flexibility does not remove the need for a strong linear comparison. Indeed, the Ridge comparison includes both a pooled specification and one model per delivery hour, following the hour-specific structure common to strong linear electricity-price benchmarks [13]. It remains deliberately smaller than full LEAR because the available feature set omits load, fuel, and neighboring-market lags.

2.3 Evaluation and uncertainty

Forecast comparison should distinguish the magnitude of a sample difference from evidence that the difference is persistent. Diebold and Mariano [2] formulate tests based on loss differentials. Because the twenty-four prices of a day are jointly determined and adjacent days remain dependent, this study first aggregates absolute error by delivery day, then uses a Newey–West long-run variance estimate [15]. Confidence intervals for MAE use seven-day moving blocks of daily summaries [12], rather than treating adjacent days as independent. Forecast comparisons share information sets and the weekly naive prediction also enters the larger models as a feature. Standard equal-accuracy asymptotics are therefore interpreted cautiously in light of nested-model results [1]; the reported daily test applies the Harvey–Leybourne–Newbold finite-sample correction [8].

Point forecasts alone do not describe operational risk. Conformal methods can wrap a point predictor with residual-based intervals, but standard finite-sample coverage relies on exchangeability. Electricity prices are a dependent, nonstationary time series. Methods designed for dynamic sequences make this limitation explicit [20]. The interval in this paper is therefore described as *prequential*: it uses only earlier residuals and reports empirical coverage. It is not presented as an exact distribution-free guarantee under arbitrary temporal dependence. Adaptive conformal methods provide a natural extension under distribution shift [6].

3 Data and information-set audit

3.1 Sources and coverage

Table 1 summarizes the frozen inputs. Prices and day-ahead renewable forecasts are obtained through the Fraunhofer ISE Energy-Charts API [4]. The API explicitly exposes DE-LU day-ahead prices and a public-power-forecast endpoint; the ingestion request fixes `forecast_type=day-ahead`, and the archived response is treated as the TSO day-ahead product named by that endpoint. This source label is a necessary information-vintage assumption, not an independently observed issue timestamp. Fifteen-minute prices, when present, are averaged to an hourly equivalent. Wind-onshore and wind-offshore forecasts are summed. Solar remains separate and is also included in total forecast renewables.

The frozen table covers 2024-08-31 through 2026-06-11, contains 15,576 observations, and has a SHA-256 fingerprint recorded in `reports/research_metrics.json`. The full-year out-of-sample window contains exactly 8,760 hours across 365 local delivery days. The first 168 feature rows are unavailable by construction because the weekly price lag has not yet formed.

3.2 Weather-vintage boundary

Open-Meteo documents its Historical Forecast API as a continuous hourly series formed by stitching the first hours of successive operational model runs [16]. That product is useful for reconstructing model-based historical conditions, but it does not fix the forecast lead time at D-1 for each delivery hour. A value close to delivery may therefore incorporate a later model run than the day-ahead auction information set.

This distinction changes the design. Temperature, 100-m wind speed, solar radiation, heating-degree, and cooling-degree variables are excluded from the primary seventeen-feature specification. A twenty-two-feature weather-augmented model is reported only as a sensitivity. The correction is conservative: the sensitivity lowers MAE by merely -0.81%. A future version can use the Open-Meteo Previous Runs or Single Runs interfaces at a fixed lead time and then reconsider weather as a primary covariate.

Remark 1 (Claim boundary). *The term “D-1 model” in the remainder of the paper refers only to the seventeen-feature specification based on price history, TSO day-ahead renewable forecasts, and deterministic calendar variables. It does not refer to the weather-augmented sensitivity.*

3.3 Quality assurance and descriptive structure

The QA program verifies the complete expected UTC grid, duplicate timestamps, per-column null shares, physical ranges, extreme price changes, constant-value runs, and local daylight-saving day lengths. The frozen dataset has zero missing hours, zero duplicate timestamps, no declared range violation, and no local day outside 23/24/25 hours after excluding partial endpoints. Two weather observations are initially missing. The primary sample is formed by dropping missing values only from the seventeen primary features and target; weather completeness is imposed separately for the weather sensitivity. Consequently, optional weather availability cannot condition the primary estimation sample. In this frozen dataset the two samples nevertheless retain the same 8,760 out-of-sample rows.

Table 1: Frozen data used by the computational artifact. The primary model uses the first three rows only.

Series	Source and unit	Role and treatment
DE-LU day-ahead price	Energy-Charts; EUR/MWh	Target; 15-minute values averaged to hourly equivalents; lags of 24, 48, and 168 hours
Solar forecast	Energy-Charts; MW	Day-ahead TSO forecast; primary exogenous feature
Wind onshore/offshore forecast	Energy-Charts; MW	Day-ahead TSO forecast; summed to total wind; primary exogenous feature
Temperature, 100-m wind, radiation	Open-Meteo; physical units	Stitched historical forecast; excluded from the primary claim and retained only for sensitivity

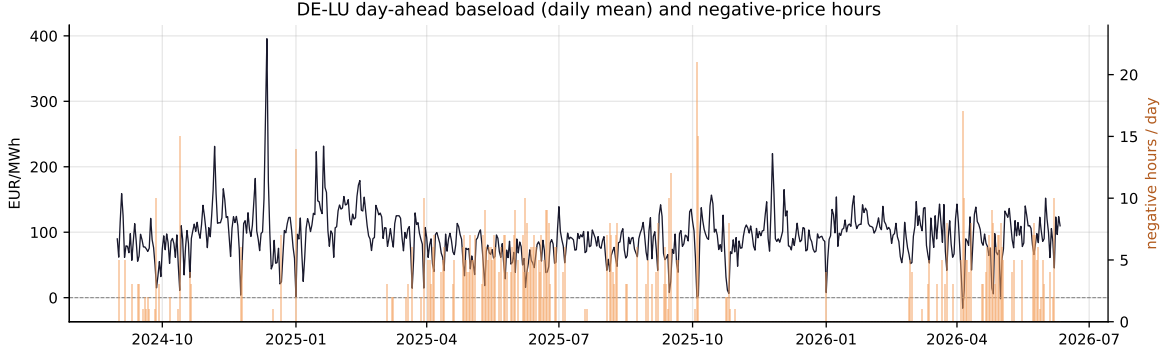


Figure 1: DE-LU day-ahead daily mean price and the number of negative-price hours. Hourly observations are retained for every model and metric.

Figure 1 shows daily mean prices and the count of negative hours. The sample includes calm and volatile periods, but it is not a multi-cycle historical panel. Its brevity is a central limitation rather than a detail to be hidden.

4 Forecasting design

4.1 Features

For target hour t in delivery day d , the primary feature vector is

$$x_t = (y_{t-24}, y_{t-48}, y_{t-168}, \bar{y}_t^{(d-1)}, s_t^{(d-1)}, \bar{y}_t^{(7d)}, S_t, W_t, R_t, \Delta_{24}R_t, C_t),$$

where S_t and W_t are day-ahead solar and total-wind forecasts, $R_t = S_t + W_t$, and C_t contains hour, weekday, month, weekend, holiday, and sine/cosine hour encodings. The rolling price statistics are shifted by at least 24 hours before aggregation. Thus no realized price from delivery day d is used to forecast that day.

The day-over-day renewable change,

$$\Delta_{24}R_t = R_t - R_{t-24},$$

provides a simple ramp indicator. It is an input forecast difference, not a realized generation change. All continuous quantities are left in natural units for the tree model. Ridge is fitted inside a standardization pipeline so its penalty does not depend on units. The integer hour, weekday, and month fields are intentionally available to LightGBM as ordered split variables; cyclic sine/cosine hour terms represent the midnight boundary. This compact treatment avoids estimating sparse

categories in the short initial window, but it restricts calendar partitions to contiguous integer groups and is therefore listed as model risk.

4.2 Models

The weekly naive forecast is

$$\hat{y}_t^N = y_{t-168}.$$

It preserves the same weekday-hour shape and is difficult to beat only when the market changes little week to week. Its weakness is deliberate: it provides a transparent reference, not a straw-man claim that all persistence models are equivalent.

Ridge estimates

$$\hat{\beta}_\lambda = \arg \min_{\beta} \sum_{t \in \mathcal{T}} (y_t - \beta_0 - x_t^\top \beta)^2 + \lambda \|\beta\|_2^2, \quad \lambda = 10.$$

The penalty is fixed before the reported out-of-sample window. No hyperparameter selection uses test losses.

The stronger linear benchmark fits the same standardized Ridge specification separately for each of the 24 local delivery-hour labels. DST duplicate hours share the same hour-specific model. This adds intraday coefficient flexibility without using test outcomes or introducing a hyperparameter search. It is LEAR-like in hourly structure, but it is not the full high-dimensional LEAR benchmark of [13].

LightGBM estimates an additive tree model

$$\hat{f}_M(x) = \sum_{m=1}^M \nu h_m(x),$$

with $M = 600$, learning rate $\nu = 0.05$, 63 leaves, minimum leaf sample size 40, row subsampling 0.9, column subsampling 0.9, and ℓ_2 regularization 1. The random seed is 42. Deterministic column-wise training is requested explicitly. These values are frozen; the paper does not report a search over alternative hyperparameters.

4.3 Expanding-window protocol

Let $D_1, \dots, D_{365}$ be the final 365 local delivery days. For each D_i , the training set contains every feature row with local date strictly less than D_i . Models are re-estimated when $i \equiv 1 \pmod{7}$ and reused for the next six delivery days. The entire day, including a DST transition, is predicted as one test block.

The design is intentionally test-heavy: 365 local days are reserved for evaluation, while the first forecast is trained on 277 complete feature days. The initial model has not seen a full prior annual cycle. To expose possible cold-start effects, the paper also reports LightGBM MAE after excluding the first 90 test days.

Proposition 1 (No target leakage through the split). *If each primary feature for delivery day d is measurable before the $D-1$ auction and the training mask is $\{t : \text{day}(t) < d\}$, then no realized price from day d enters either the fitted parameters or the feature vector used to forecast day d .*

Proof. The training mask excludes every timestamp with local delivery date d . Each rolling price feature is shifted by at least 24 hours, and the remaining primary features are a deterministic calendar or a day-ahead TSO forecast. Therefore both the fitted sample and x_t for $t \in d$ are measurable before the target prices y_t are realized. $\square$

This proposition is conditional on the source labels and the implementation. It does not establish that an upstream archive never revised a historical record. That remaining data-vintage risk is discussed in Section 8.

4.4 Point metrics

For errors $e_t = \hat{y}_t - y_t$ over n hours,

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |e_t|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}, \quad \text{Bias} = \frac{1}{n} \sum_{t=1}^n e_t.$$

MAE is primary because it remains interpretable with negative prices and does not square the already influential spikes. RMSE and the 95th percentile of absolute error retain information about tails.

For uncertainty in MAE, the code first computes a mean absolute error for each local delivery day and resamples overlapping seven-day blocks of the 365 daily values 4,000 times [12]. The 2.5 and 97.5 percentiles form the reported interval. This preserves within-day dependence and short inter-day dependence aligned with the HAC lag; it remains unable to reproduce arbitrary long regime shifts.

4.5 Loss-differential test

For comparator A and candidate B , define daily absolute loss

$$L_{d,m} = \frac{1}{H_d} \sum_{h=1}^{H_d} |y_{d,h} - \hat{y}_{d,h}^m|, \quad \delta_d = L_{d,A} - L_{d,B}.$$

Positive δ_d favors B . The sample mean $\bar{\delta}$ is divided by a Newey–West standard error with Bartlett weights and lag seven:

$$\hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{k=1}^7 \left(1 - \frac{k}{8}\right) \hat{\gamma}_k, \quad t = \frac{\bar{\delta}}{\sqrt{\hat{\Omega}/n_d}}.$$

The statistic is multiplied by the Harvey–Leybourne–Newbold small-sample factor for a one-day forecast horizon [8]. This remains a Diebold–Mariano-style comparison on daily L_1 loss. The weekly naive forecast is also an input feature, so that comparison shares the nested-model problem studied by [1]; Clark–West’s squared-error adjustment does not transfer directly to the primary absolute-error loss. The fixed HAC lag and reference distribution remain approximations, and p -values are strength-of-evidence diagnostics rather than exact finite-sample tests.

4.6 Prequential intervals

Let $a_t = |y_t - \hat{y}_t|$ be the absolute residual. Before predicting delivery day d , the interval procedure takes residuals from the previous 60 delivery days, provided at least 30 days exist. With n_c calibration residuals and $\alpha = 0.10$, it sets

$$q_d = Q_{\lceil (n_c+1)(1-\alpha) \rceil / n_c} (\{a_t : \text{day}(t) < d\}),$$

using the higher empirical quantile, and reports

$$C_{d,h} = [\hat{y}_{d,h} - q_d, \hat{y}_{d,h} + q_d].$$

The coefficients of the point model are not refitted by the interval wrapper. The construction is strictly prequential in residual time but pools all hours into one radius. Hour-specific coverage therefore remains an essential diagnostic. Alongside coverage and width, the artifact reports the proper central interval score and two endpoint pinball losses [7]. Hour-by-hour two-sided binomial tests are reported both unadjusted and with Holm multiplicity correction.

4.7 Prompt-proxy translation

The model’s next-day baseload fair value is

$$F_d = \frac{1}{H_d} \sum_h \hat{y}_{d,h}.$$

Because licensed EEX front-week settlements are not distributed with the artifact, the prompt proxy is the trailing w -day realized day-ahead baseload,

$$P_{d,w} = \frac{1}{w} \sum_{j=1}^w \bar{y}_{d-j}.$$

The gap $G_{d,w} = F_d - P_{d,w}$ is standardized using a trailing 60-day mean and standard deviation. A long direction is recorded when $z_{d,w} > \tau$, a short direction when $z_{d,w} < -\tau$, and neutral otherwise.

For direction $s_d \in \{-1, 0, 1\}$, the descriptive outcome is

$$R_d = s_d \left(\frac{1}{7} \sum_{j=0}^6 \bar{y}_{d+j} - P_{d,w} \right).$$

The seven-day forward average creates overlapping outcomes. A fourteen-day moving-block bootstrap [12] is therefore used for the default $(w, \tau) = (7, 0.75)$ summary. The final six days have no complete forward outcome and are right-censored. They remain available for a current view but are excluded from performance statistics.

5 Empirical results

5.1 Point accuracy

Table 2 reports the full-year comparison. LightGBM reaches an MAE of 13.59 EUR/MWh, against 15.73 for hourly Ridge, 17.11 for pooled Ridge, and 32.89 for the weekly naive forecast. The primary model’s RMSE is 22.22, larger than MAE as expected in a spiky market, and its bias is -3.72 EUR/MWh. The negative bias indicates systematic underprediction on average, a warning that the model should not be judged on MAE alone.

After excluding the first 90 test days, LightGBM MAE over the remaining 275 days is 14.11 EUR/MWh. This diagnostic separates the short initial training history from the headline full-year score without silently discarding a difficult part of the declared test.

A strictly prequential intercept correction is also reported as a sensitivity, not substituted into the primary forecast. On the common post-calibration window, the uncorrected MAE and bias are 13.42 and -3.29 EUR/MWh; subtracting the prior-60-day mean residual changes them to 13.44 and 0.17. This quantifies the available bias correction while avoiding an ex-post full-sample intercept.

Figure 3 illustrates the last fourteen delivery days. The nonlinear forecast tracks daily shape more closely than the weekly naive profile, but misses several sharp ramps. Figure 4 gives the full OOS scatter. Compression toward the center is visible: high realized prices are underpredicted and some low prices are overpredicted.

5.2 Statistical comparison and temporal stability

The daily MAE gain is 19.29 EUR/MWh relative to the naive model, 3.52 relative to pooled Ridge, and 2.13 relative to hourly Ridge. All HLN-corrected, autocorrelation-robust tests reject a zero mean differential by a wide margin (Table 3). These results support the within-sample statement that the gain is persistent across days, rather than being generated by one small group of hours.

Figure 5 shows that the expanding gain remains positive throughout the year for both comparisons. It should not be extrapolated mechanically: the data include one recent annual cycle, not repeated crises, fuel regimes, or structural policy changes.

Table 2: One-year expanding-window point-forecast accuracy. MAE intervals use 4,000 seven-day moving-block bootstrap draws of delivery-day summaries.

Model	MAE [95% CI]	RMSE	Bias	Median AE	95th pct. AE
Weekly naive	32.89 [29.23, 36.74]	51.21	-0.87	19.27	102.48
Pooled Ridge	17.11 [16.04, 18.26]	25.41	-3.23	13.24	42.18
Hourly Ridge (24 models)	15.73 [14.78, 16.75]	23.48	-3.32	12.22	39.19
LightGBM	13.59 [12.39, 14.86]	22.22	-3.72	9.09	38.61

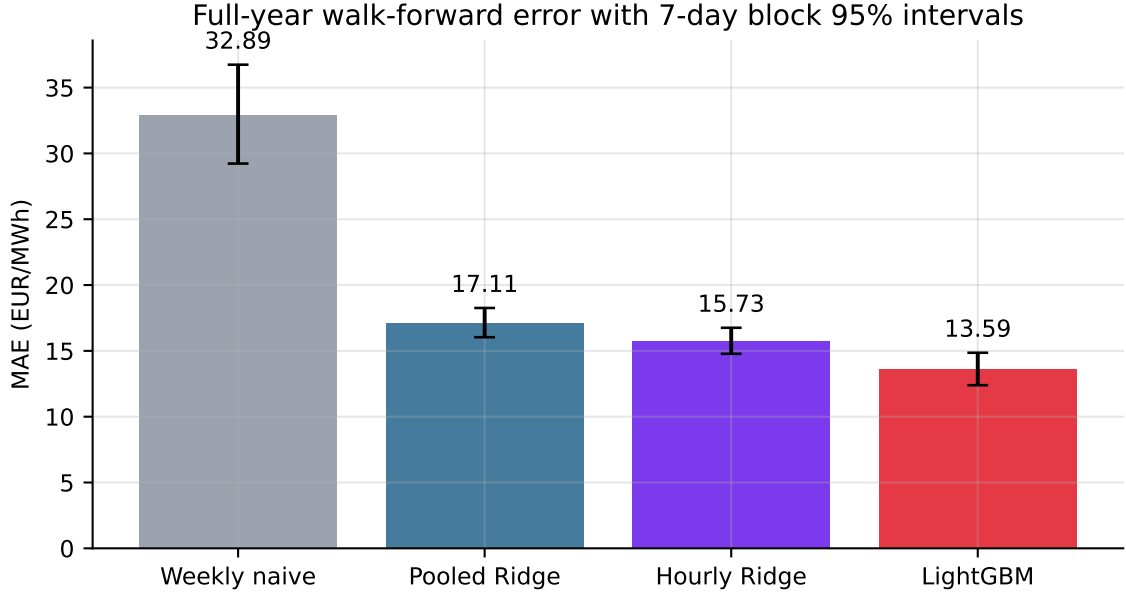


Figure 2: MAE and seven-day moving-block bootstrap 95% intervals. Intervals describe variability across this sample, not model or data-vintage risk.

5.3 Hour, season, and feature importance

Errors are not uniform through the day. Figure 6 shows that night and late-evening hours are easier, while daylight and the evening ramp are harder. The nonlinear model’s MAE is 9.30 EUR/MWh during hours 00–06, 16.65 during hours 07–15, 15.49 during hours 16–21, and 9.16 during hours 22–23. Spring has the largest seasonal MAE at 16.30, followed by autumn at 14.20, summer at 11.93, and winter at 11.92.

Split-count importance is reported as an implementation diagnostic, not a causal attribution. The most frequently selected variables are the 24-hour price lag, trailing weekly price mean, total renewable forecast, prior-day price dispersion, prior-day mean, and day-over-day renewable change. Correlated features can substitute for one another, so neither rank nor split count measures economic marginal effect.

5.4 Ablation evidence

Table 5 and Figure 9 keep the model class, dates, refit cadence, and hyperparameters fixed while changing feature families. The weather-augmented sensitivity improves MAE by only -0.81% relative to the primary model. This small difference is inside the broad moving-block bootstrap intervals and does not justify relaxing the information-vintage standard.

Renewables are materially more important. Removing them leaves price history and calendar variables and raises MAE by 84.9%. A model with renewables and calendar variables but no lagged prices obtains MAE 17.80, which remains substantially better than price-history plus calendar

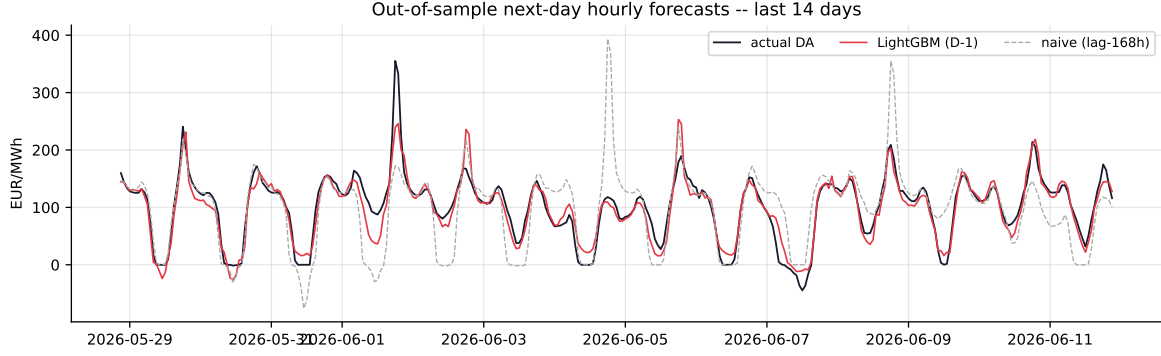


Figure 3: Hourly out-of-sample forecasts for the final fourteen days.

Table 3: Daily absolute-loss comparisons with seven-lag HAC standard errors and the Harvey–Leybourne–Newbold finite-sample correction. Positive gains favor LightGBM.

Comparison	Mean daily MAE gain	HAC SE	HLN statistic	Two-sided p
LightGBM vs. Weekly naive	19.29	1.64	11.78	2.35×10^{-27}
LightGBM vs. pooled Ridge	3.52	0.37	9.61	1.23×10^{-19}
LightGBM vs. hourly Ridge	2.13	0.37	5.82	1.28×10^{-8}

at 25.14. The result suggests complementarity: recent prices establish the prevailing level while day-ahead renewable forecasts explain a large fraction of the next-day deviation.

5.5 Tail behavior

Aggregate improvement coexists with economically relevant failure modes. Table 6 divides the target into negative, regular, and above-200 EUR/MWh hours. On regular hours, LightGBM obtains MAE 12.14; on negative hours, its MAE is 21.59; above 200, MAE is 61.12. The high-price bias is strongly negative, reflecting regression toward the center. The model predicts a value above 200 for only 53.5% of realized above-200 hours.

Ridge slightly outperforms LightGBM on negative-price hours in MAE, while LightGBM is materially better in the high-price regime. This is a useful counterexample to any claim of uniform dominance. A production forecaster might combine models by regime, use a transformed target, add outages and fuel-stack variables, or train an explicit tail objective. Variance-stabilizing transformations designed for zero and negative electricity prices are a particularly relevant extension for these documented tails [18].

5.6 Prequential interval performance

Across 8,040 evaluable hours after the minimum calibration period, the nominal 90% interval covers 90.07% of outcomes with mean width 59.03 EUR/MWh and mean proper interval score 97.77. Aggregate calibration is close to the target. Figure 11 illustrates how the radius adapts through the last fourteen days.

The aggregate number conceals a formal conditional-coverage failure. 18 of the 24 hourly coverages differ from 90% under two-sided normal tests at 5%: 9 under-cover and 9 over-cover. After Holm correction across the 24 tests, 16 remain significant. A common radius is too wide in easy hours and too narrow in difficult hours. Hour-conditional residual pools, quantile regression, or an adaptive conformal method [6] are natural extensions. Because the time series is nonexchangeable, the marginal empirical rate remains a descriptive validation result rather than a conditional or finite-sample coverage theorem.

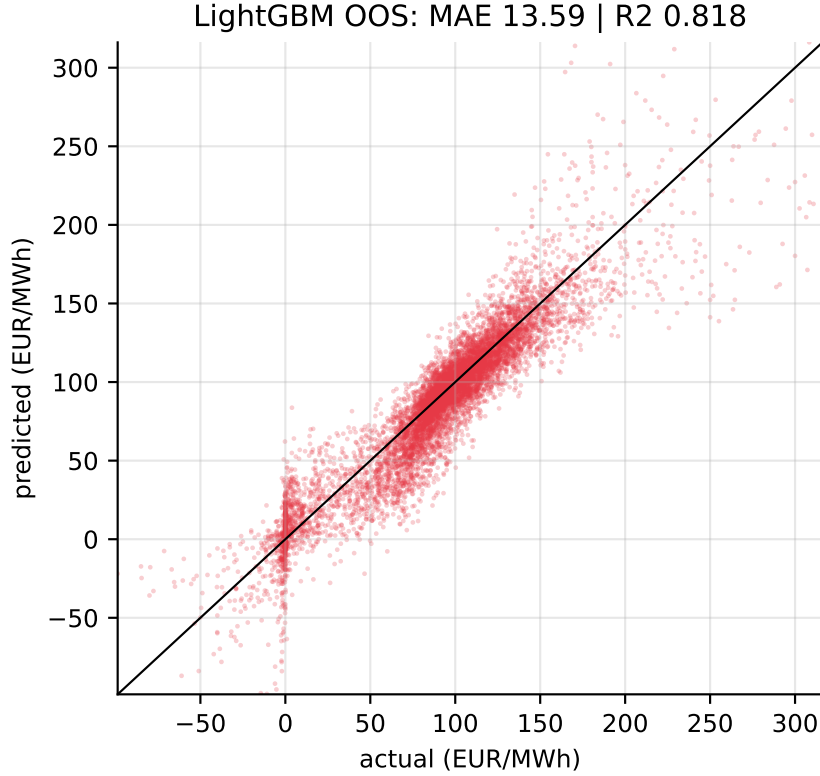


Figure 4: Observed versus LightGBM prices for all 8,760 out-of-sample hours.

Table 4: Primary-model errors by broad season and hour block.

Dimension	Regime	n	MAE	RMSE	Bias
Season	Autumn	2,185	14.20	23.29	-5.04
Season	Spring	2,207	16.30	28.08	-6.02
Season	Summer	2,208	11.93	18.70	-3.25
Season	Winter	2,160	11.92	17.03	-0.50
Hour Block	Daylight (07–15)	3,285	16.65	27.29	-4.38
Hour Block	Evening ramp (16–21)	2,190	15.49	24.74	-4.24
Hour Block	Late evening (22–23)	730	9.16	12.73	-2.40
Hour Block	Night (00–06)	2,555	9.30	12.80	-2.78

5.7 Prompt-proxy diagnostic

Figure 13 plots the next-day baseload fair value, the trailing seven-day day-ahead prompt proxy, and the standardized gaps. At the declared $|z| > 0.75$ threshold, 144 active observations have complete forward-week outcomes. The directional hit rate is 74.3% and the mean signed spread is 13.25 EUR/MWh. A fourteen-day moving-block bootstrap gives a hit-rate interval of [68.2%, 80.3%] and a mean-spread interval of [9.55, 17.66] EUR/MWh.

Table 7 bounds the raw result. Always-long and weekly naive references quantify simple persistence. Pooled and hourly Ridge show how much of the direction survives a materially weaker point forecaster, while perfect knowledge of day-D baseload is an infeasible upper bound for this particular construction. LightGBM carries direction beyond weekly persistence, but the proximity of the linear and perfect-information rows shows that much of the headline hit rate is mechanical: day D enters the forward-week average and the prompt spread is autocorrelated. This is evidence of directional information, not model-specific alpha.

Sensitivity matters. Raising the threshold generally increases the observed hit rate and magnitude while sharply reducing the sample. The best-looking ex-post cell is therefore not evidence for

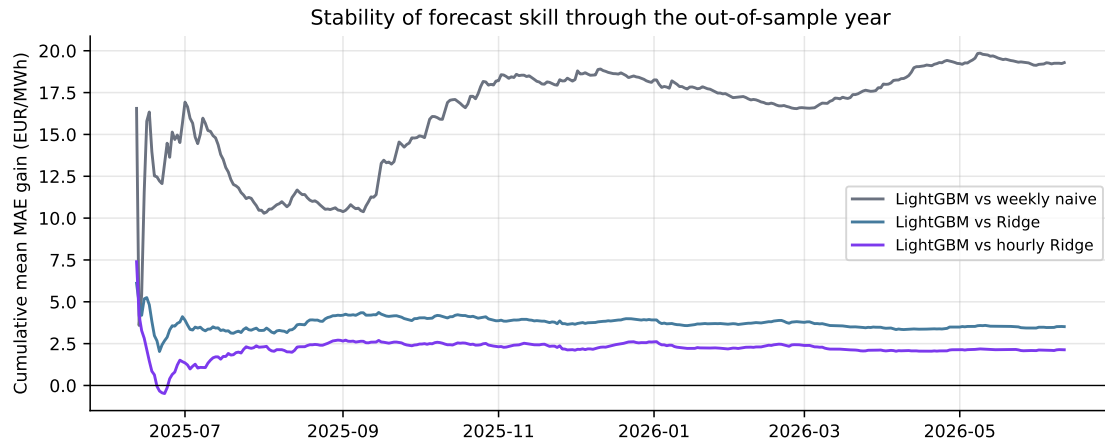


Figure 5: Expanding average daily MAE gain of LightGBM. A line above zero means that the comparator has incurred greater cumulative error.

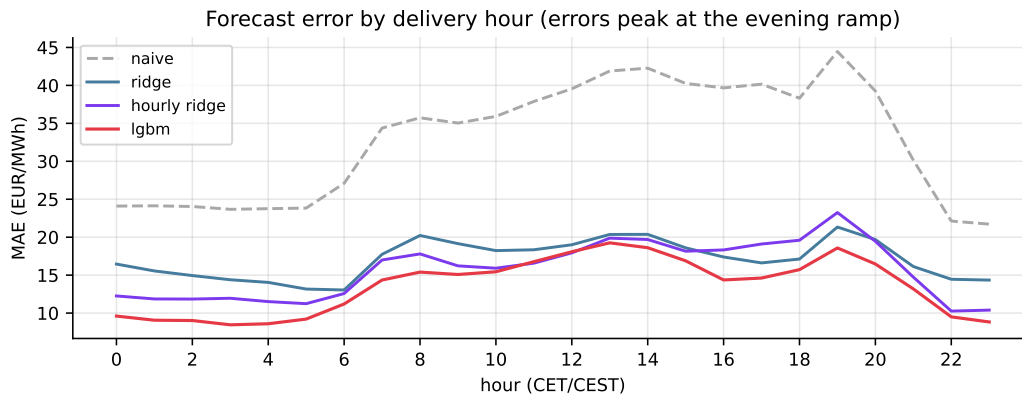


Figure 6: MAE by local delivery hour for all four models.

an optimized live threshold. The value 0.75 is retained because it was declared in the prototype before this expanded analysis.

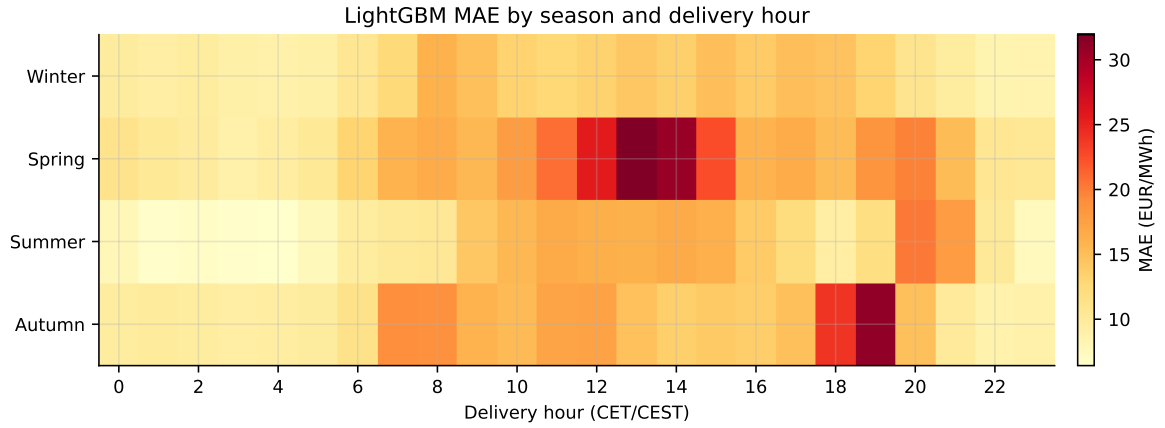


Figure 7: LightGBM MAE across season-hour cells. The heat map makes clear that aggregate calibration does not imply uniform conditional performance.

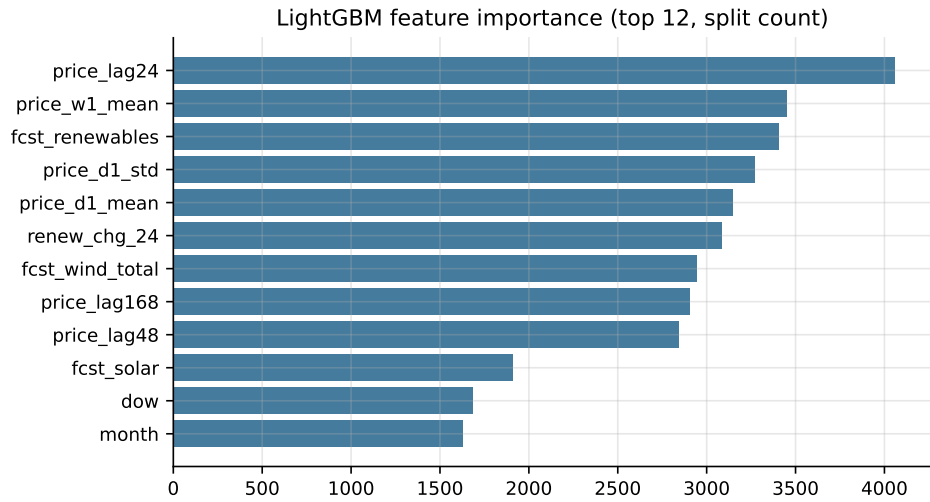


Figure 8: Average LightGBM split-count importance across the weekly refits.

Table 5: Feature-family ablations. The weather-augmented row is a sensitivity, not part of the primary ex-ante claim.

Specification	Features	MAE	RMSE	Bias	MAE vs. primary
Weather-augmented sensitivity	22	13.49	21.85	-3.59	-0.8%
Primary D-1 specification	17	13.59	22.22	-3.72	+0.0%
Renewables + calendar	11	17.80	26.32	-6.34	+30.9%
Price history + calendar	13	25.14	37.14	-0.84	+84.9%
Calendar only	7	28.89	40.69	+0.30	+112.5%

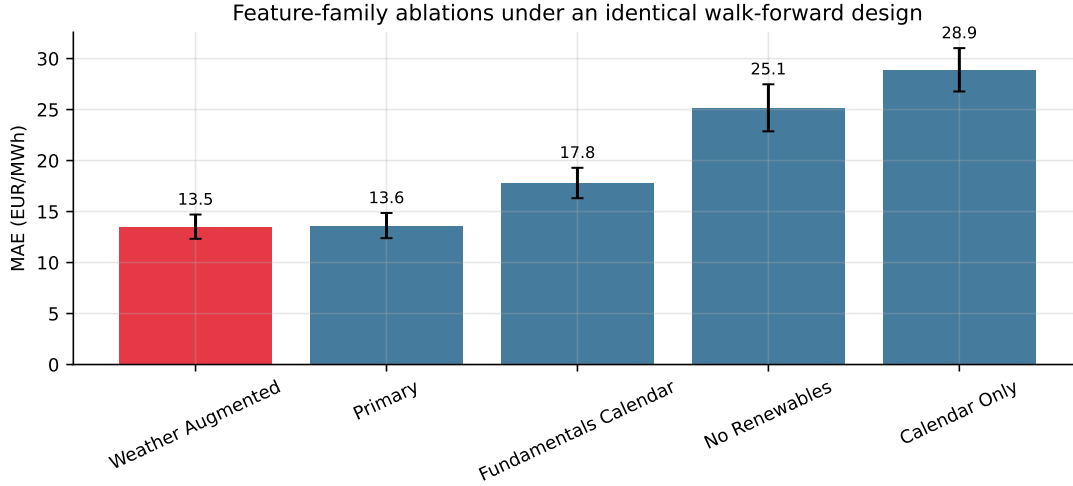


Figure 9: Ablation MAE with delivery-day bootstrap intervals.

Table 6: Errors conditional on realized price regime. Conditioning on the outcome is diagnostic and not an ex-ante trading rule.

Price regime (EUR/MWh)	Model	n	MAE	Bias
Price < 0	Weekly naive	538	55.22	+42.39
Price < 0	Pooled Ridge	538	21.47	+0.99
Price < 0	Hourly Ridge	538	21.51	-1.90
Price < 0	LightGBM	538	21.59	+1.41
$0 \leq \text{price} < 200$	Weekly naive	8,065	29.74	-1.60
$0 \leq \text{price} < 200$	Pooled Ridge	8,065	15.69	-2.18
$0 \leq \text{price} < 200$	Hourly Ridge	8,065	14.34	-2.22
$0 \leq \text{price} < 200$	LightGBM	8,065	12.14	-3.33
Price ≥ 200	Weekly naive	157	118.34	-111.46
Price ≥ 200	Pooled Ridge	157	74.99	-71.42
Price ≥ 200	Hourly Ridge	157	67.07	-64.58
Price ≥ 200	LightGBM	157	61.12	-40.96

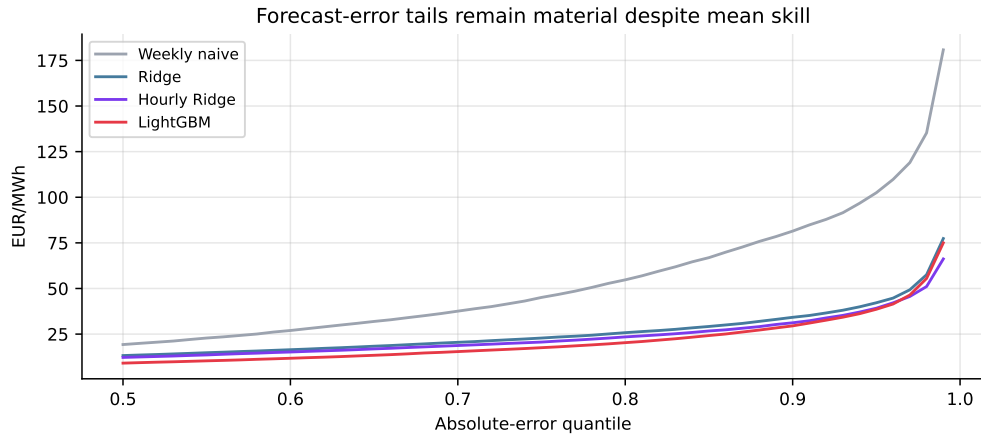


Figure 10: Absolute-error quantile functions. The nonlinear model's advantage narrows in the far tail.

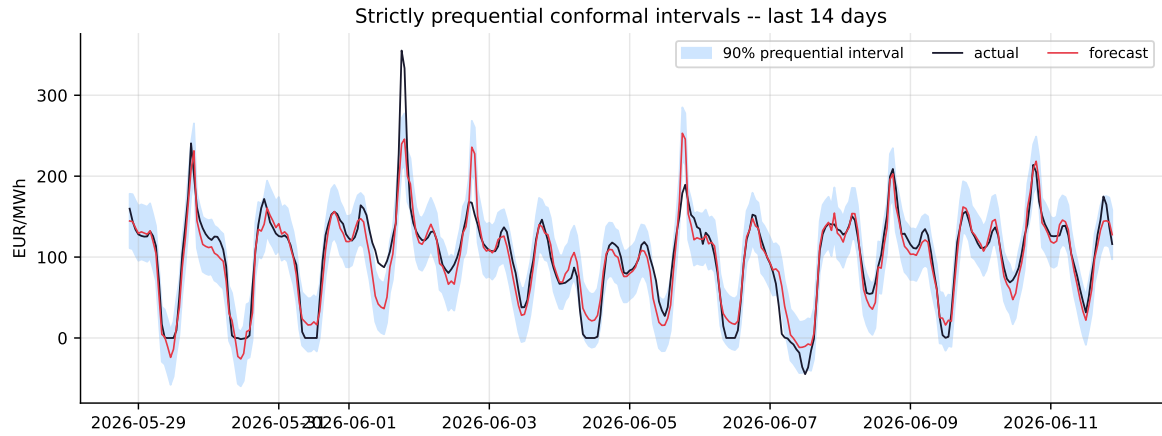


Figure 11: Forecasts and rolling-residual 90% intervals over the final fourteen days.

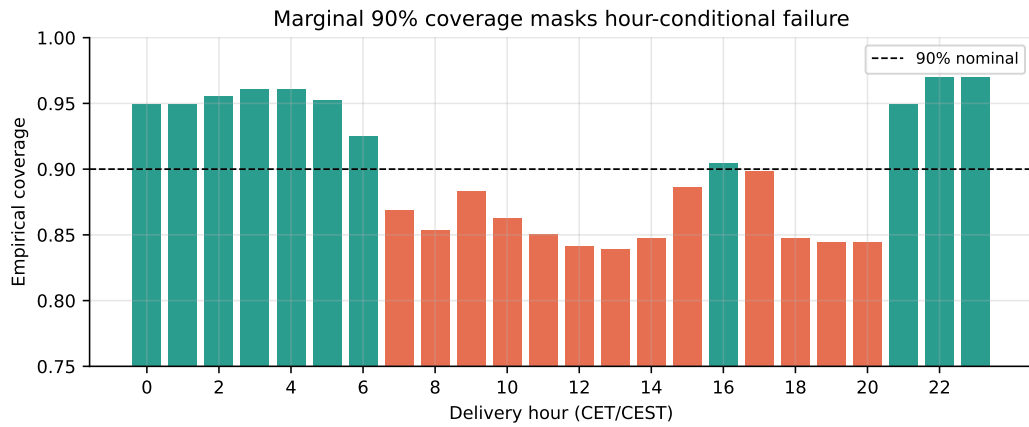


Figure 12: Empirical interval coverage by local delivery hour. Marginal calibration masks widespread conditional rejection of the 90% target.

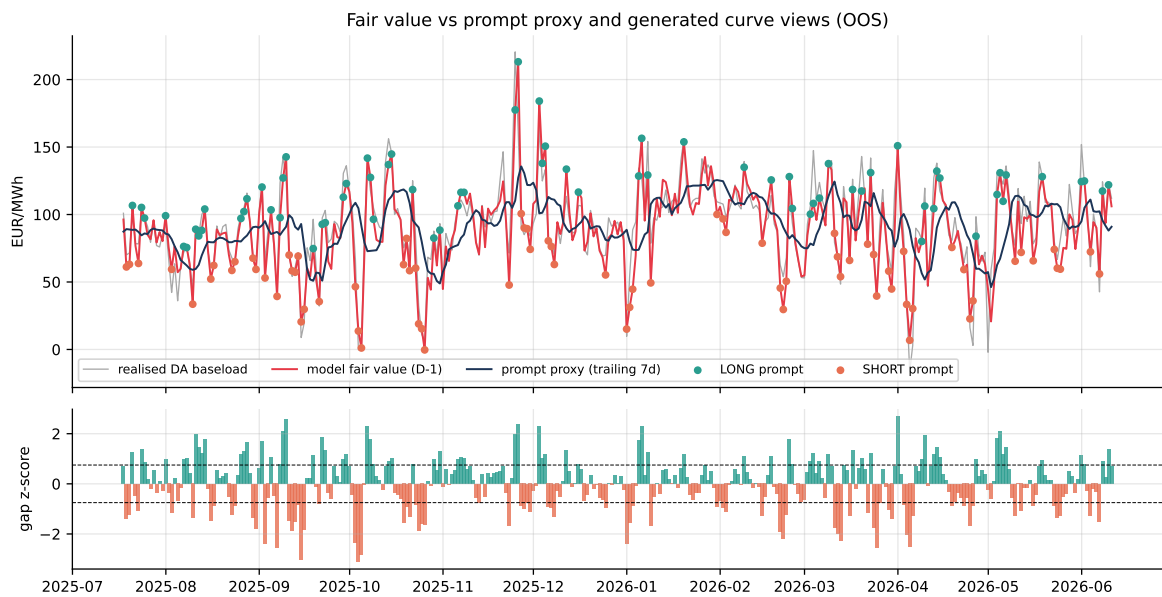


Figure 13: Next-day baseload fair value, trailing day-ahead prompt proxy, and generated directions. This is not an EEX forward backtest.

Table 7: Bounds for the declared seven-day, $|z| > 0.75$ prompt diagnostic. Mean spread is signed EUR/MWh. “Perfect day-D” is deliberately infeasible.

Signal input	Feasible	Active days	Hit rate	Mean spread
Always long (no model signal)	yes	352	48.3%	0.55
Weekly naive fair value	yes	121	54.5%	1.71
Pooled Ridge fair value	yes	148	71.6%	12.44
Hourly Ridge fair value	yes	150	73.3%	13.32
LightGBM fair value	yes	144	74.3%	13.25
Perfect day-D baseload (infeasible)	no	139	81.3%	14.96

Table 8: Threshold sensitivity for the seven-day prompt proxy. Mean and median columns are signed forward-week spreads in EUR/MWh.

$ z $ threshold	Active days	Hit rate	Mean spread	Median spread	Long/short
0.50	200	71.5%	11.15	10.50	102/98
0.75	144	74.3%	13.25	12.34	70/74
1.00	105	79.0%	13.78	11.99	52/53
1.25	72	83.3%	17.41	18.05	29/43
1.50	47	80.9%	18.30	22.18	19/28
2.00	21	76.2%	13.71	15.39	8/13

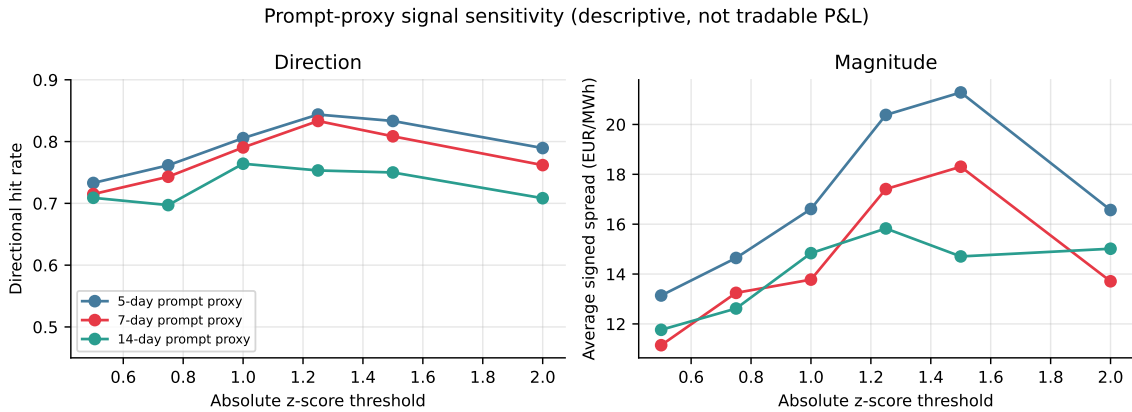


Figure 14: Signal sensitivity across prompt windows and thresholds. The outcomes overlap and are descriptive only.

6 Discussion

6.1 What the evidence supports

The strongest conclusion is comparative and local. Under one fixed implementation, a complete OOS year, and the declared D-1 feature set, LightGBM materially reduces absolute error relative to a weekly naive model and both Ridge specifications. The improvement appears through the year, survives daily dependence adjustments, and is not explained by the ambiguous-vintage weather series. Renewable forecasts are a central source of skill.

The artifact also shows why an MAE headline is insufficient. The primary model is negatively biased, its daytime interval coverage is weak, and its error increases sharply in negative and high-price regimes. Ridge is competitive on negative prices. These facts are not ancillary; they define the operational boundary of the model. Likewise, the prompt diagnostic exceeds weekly persistence but is largely shaped by its overlapping arithmetic rather than unique LightGBM alpha.

6.2 What the evidence does not support

The results do not establish causal effects of renewable generation on prices. The feature importance and ablations measure predictive contribution inside this design. Renewables correlate with demand, outages, fuel costs, and cross-border conditions that are not fully observed.

The results do not establish state-of-the-art performance. The repository does not compare with LEAR, neural networks, probabilistic gradient boosting, or commercial forecast vendors. It also does not tune LightGBM against the test year. This restraint makes the comparison easier to audit but narrower than a comprehensive benchmark.

Finally, the prompt-proxy output is not a backtested financial return. Licensed front-week settlements, bid-ask spreads, transaction costs, margin, execution timing, and risk limits are absent. A trailing day-ahead average does not contain the forward risk premium. The experiment measures whether a standardized fair-value divergence aligns with subsequent day-ahead baseload, nothing more.

6.3 Operational interpretation

A practical forecasting desk could use the point model as one component of a larger process. The next-day curve provides an hourly shape and a baseload summary. The prequential interval gives a first uncertainty band, while the hourly coverage chart identifies where that band is least trustworthy. Invalidation rules should include major TSO forecast revisions, REMIT outages, fuel and carbon shocks, and an error beyond a rolling tolerance.

The current implementation deliberately keeps those rules outside the statistical score. Without timestamped outage, fuel, carbon, and actual forward data, coding them into a historical strategy would create an illusion of precision.

7 Software artifact and reproducibility

7.1 Architecture

The repository is divided into seven executable stages:

1. `ingest.py` downloads and caches source responses;
2. `qa.py` validates the assembled hourly dataset;
3. `features.py` constructs the declared information set;
4. `models.py` runs the full-year walk-forward experiment;
5. `trading.py` builds the right-censoring-aware prompt proxy;

6. `research.py` produces uncertainty, ablation, regime, and sensitivity evidence;
7. `make_figures.py` and `generate_paper_artifacts.py` render figures, tables, and macros directly from the frozen outputs.

Tabular transformations use pandas [14]; standardized Ridge, metrics, and preprocessing use scikit-learn [17]. These software dependencies are pinned in the paper environment.

The optional language-model component writes a short desk note from pipeline-computed numbers. It is excluded from every scientific result. Its dry-run mode logs the exact prompt without credentials or a network call. No paper table or figure depends on generated prose.

7.2 Evidence lineage

Table 9: Lineage from numerical claim to executable evidence.

Claim	Producer	Frozen artifact
Point accuracy and intervals	<code>models.py</code> , <code>research.py</code>	<code>model_comparison.csv</code> , <code>model_daily_losses.csv</code>
Loss-differential significance	<code>research.py</code>	<code>dm_tests.csv</code>
Feature-family contribution	<code>research.py</code>	<code>ablation_metrics.csv</code> , <code>ablation_predictions.csv</code>
Regime and tail errors	<code>research.py</code>	<code>regime_metrics.csv</code> , <code>tail_metrics.csv</code>
Prequential coverage	<code>research.py</code>	<code>conformal_diagnostics.csv</code> , <code>conformal_hour_tests.csv</code>
Bias sensitivity	<code>research.py</code>	<code>bias_correction_diagnostics.csv</code>
Prompt-proxy sensitivity and bounds	<code>trading.py</code> , <code>research.py</code>	<code>daily_views.csv</code> , <code>signal_sensitivity.csv</code> , <code>signal_benchmarks.csv</code>

Every figure is generated in both PNG and vector PDF form. The README uses PNGs; this paper uses the matching PDFs. Generated LaTeX tables are not manually transcribed. Dataset and prediction hashes provide a quick audit that a reported paper was built from the intended frozen state.

7.3 Tests and deterministic controls

The test suite checks price-lag direction, independent primary/weather sample conditioning, QA invariants, the HLN/HAC sign convention, seeded moving-block resampling, strictly prior conformal calibration, the exact 365-day OOS shape, all four forecast columns, prompt right-censoring and benchmark bounds, equality between primary ablation and frozen model predictions, conditional coverage reconciliation, dataset hashing, and the presence of all fourteen vector figures.

LightGBM uses an explicit seed and deterministic training mode. Bootstrap generators use fixed seeds. Exact floating-point equality across operating systems is not promised: compiler, math-library, and LightGBM versions can change last-bit results. The locked paper environment and committed CSV evidence make such differences visible.

8 Limitations and model risk

1. **Sample length.** The dataset spans roughly twenty-one months and the test covers one year. Lago et al. [13] advocate longer multi-year evaluation. This artifact does not cover multiple energy crises, policy regimes, or fuel cycles. More than half of the raw local days are reserved for testing, leaving only 277 complete feature days before the first forecast and no full prior annual cycle.

2. **Upstream revisions.** The local code enforces chronological splits, but an upstream API may revise historical forecasts. Raw response caching protects future reruns, not the first retrieval. True vintage validation would require archive snapshots with issue timestamps.
3. **Weather.** The stitched Open-Meteo historical series is not a fixed D-1 forecast. It is excluded from the primary claim. The sensitivity must not be described as live-reproducible until fixed-lead runs are used.
4. **Fundamental omissions.** Load forecasts, outages, interconnector availability, hydro conditions, TTF gas, coal, EUA carbon, and neighboring-market forecasts are absent. These omissions are likely material during scarcity and fuel-led moves.
5. **Model selection.** Hyperparameters are fixed rather than selected in a nested time-series validation. That reduces researcher degrees of freedom but does not prove the chosen values are optimal. Calendar integers are treated as ordered tree-split variables; cyclic hour terms help, but noncontiguous weekday or month groupings are unavailable.
6. **Intervals.** The symmetric pooled conformal radius assumes neither hour-specific scale nor asymmetric tails. Aggregate coverage near 90% coexists with formal hour-conditional rejection, and temporal dependence prevents a naive exchangeability guarantee.
7. **Significance.** The HAC lag, seven-day moving-block bootstrap, HLN adjustment, and reference distribution are modeling choices. The naive forecast is nested in the feature set, while the primary loss is absolute rather than squared error. Extremely small p -values indicate strong sample evidence, not an immutable population ranking.
8. **Prompt proxy.** A trailing day-ahead mean is not a forward settlement and contains no explicit risk premium. Overlapping forward-week outcomes, mechanical persistence, missing costs, and threshold exploration preclude a trading performance claim. Similar Ridge and LightGBM hit rates show that the raw success rate is not model-specific alpha.
9. **Interpretability.** Split-count importance is unstable under correlated features and is not causal. Ablations are more informative but can also change interactions and model capacity.
10. **Production readiness.** The artifact has caching, QA, and deterministic outputs, but lacks a feature store, late-data policy, monitoring service-level objectives, access controls, and an audited execution interface.

9 Conclusion

This paper presents a reproducible DE-LU day-ahead forecasting artifact that treats information timing and uncertainty as first-class research objects. On a complete out-of-sample year, the primary LightGBM model obtains 13.59 EUR/MWh MAE and improves on a weekly naive model, pooled Ridge, and the stronger hourly Ridge baseline. The gain is stable across the sample and significant under an HLN-corrected daily autocorrelation-robust comparison. Renewable forecasts account for a large share of predictive value. The more permissive weather variant changes MAE only marginally and is estimated on its own complete-case sample because its archive does not preserve a fixed D-1 vintage.

The same evidence also limits the conclusion. Error remains large in negative and above-200 price regimes; marginal 90% interval coverage masks rejection across many delivery hours; and the sample covers one recent year with a short initial training history rather than multiple structural regimes. The prompt-proxy experiment is bounded by persistence and perfect-information references and is shown to be partly mechanical, not a forward-market backtest.

The result is therefore best viewed as a publication-ready foundation: source-backed, testable, and explicit about what remains unknown. Extending the data to multiple years, introducing fixed-

vintage weather, load, outages, fuel, carbon, and interconnector inputs, and evaluating conditional probabilistic forecasts, variance-stabilized targets, and a full LEAR implementation would turn that foundation into a broader empirical study.

A Complete primary feature dictionary

Table 10: Primary model features and their timing interpretation.

Feature	Definition and timing
price_lag24	Same local-sequence hour one day earlier; auction price already published before the current D-1 gate closure.
price_lag48	Price two days earlier.
price_lag168	Same hour one week earlier; also the naive forecast.
price_d1_mean	Mean of a 24-hour rolling window applied after a 24-hour shift.
price_d1_std	Standard deviation of the same shifted window.
price_w1_mean	Mean of the prior 168 hours after a 24-hour shift.
fcst_solar	TSO day-ahead solar generation forecast in MW.
fcst_wind_total	Sum of onshore and offshore day-ahead wind forecasts in MW.
fcst_renewables	Solar plus total wind forecast.
renew_chg_24	Current day-ahead renewable forecast minus the forecast at the same hourly position one day earlier.
hour	Europe/Berlin delivery hour.
dow	Local day of week.
month	Local calendar month.
is_weekend	Indicator for Saturday or Sunday.
is_holiday	German public-holiday indicator.
hour_sin, hour_cos	Cyclic encoding of local hour.

B Executable algorithms

B.1 Algorithm A: walk-forward point forecasts

Input: feature table, ordered local delivery days, 365-day test window, seven-day refit cadence.

1. Select the final 365 distinct Europe/Berlin delivery days.
2. For test day D_i , set the training mask to every row whose local day is strictly earlier than D_i .
3. When i is the first day of a seven-day block, fit standardized Ridge pooled across hours, 24 standardized hour-specific Ridge models, and deterministic LightGBM on the training mask.
4. Predict every 23/24/25 target hour of D_i with the same fitted models; copy the 168-hour lag as the naive prediction.
5. Append timestamp, target, and all four predictions without shuffling.
6. After the final day, compute metrics and save the exact prediction matrix to CSV.

Output: data/predictions_oos.csv and reports/model_metrics.json.

B.2 Algorithm B: prequential interval

Input: ordered point predictions and targets, nominal coverage $1 - \alpha = 0.90$, 60-day residual window, 30-day minimum.

1. Group prediction timestamps by local delivery day.

2. Before day D_i , collect absolute residuals only from days $D_{\max(1, i-60)}, \dots, D_{i-1}$.
3. Compute the finite-sample-adjusted higher empirical quantile q_i .
4. Apply the same symmetric radius to every hour of D_i .
5. After observing the day, record coverage by timestamp and hour; only then can those residuals enter a later calibration window.

Output: `data/conformal_diagnostics.csv`.

B.3 Algorithm C: feature-family ablation

1. Freeze dates, targets, LightGBM parameters, seed, and refit cadence.
2. Define the primary, weather-augmented, fundamentals-calendar, no-renewables, and calendar-only feature sets.
3. Repeat the complete walk-forward procedure separately for each set.
4. Reconcile the primary ablation predictions exactly with the frozen main-model column.
5. Compute metrics and seven-day moving-block bootstrap intervals for every set.

Output: `data/ablation_predictions.csv` and `data/ablation_metrics.csv`.

B.4 Algorithm D: prompt-proxy sensitivity

1. Average hourly LightGBM predictions and targets into local daily baseload values.
2. For prompt windows 5, 7, and 14, compute the trailing realized proxy and a 60-day rolling standardized fair-value gap.
3. For thresholds 0.5 through 2.0, assign long, short, or neutral.
4. Compute the realized D through D+6 baseload spread; mark the final six dates right-censored.
5. For the declared 7-day, 0.75 configuration, resample fourteen-day moving blocks 4,000 times.

Output: `data/daily_views.csv` and `data/signal_sensitivity.csv`.

C Additional numerical tables

C.1 Regime results

Table 4 in the main paper reports season and hour blocks. The complete machine-readable table additionally includes renewable quartiles and all four models. For the primary model, MAE is 13.82, 9.98, 12.48, and 18.09 EUR/MWh across increasing renewable quartiles. The high-renewable quartile is the most difficult, consistent with greater risk of saturation and negative-price conditions.

C.2 Threshold grid

The six rows in Table 8 should be read jointly. Higher thresholds select more extreme gaps and fewer dates. Because the same sample is used to display all cells, selecting the largest ex-post hit rate would introduce multiple-comparison bias. No row other than the predeclared 0.75 threshold is used for an uncertainty claim.

Table 11: Validation matrix for the released artifact.

Component	Property	Acceptance criterion
Dataset grid	Complete UTC hours	Zero missing and duplicate timestamps
DST mapping	Local day length	Every full day has 23, 24, or 25 hours
Feature timing	Price lags	Exact 24/48/168-hour backward mapping
Walk-forward split	Test horizon	365 local days, 8,760 rows
Linear benchmark	Hour-specific fit	24 Ridge models, refit every seven days
Primary ablation	Evidence reconciliation	Exact equality with main Light-GBM predictions
Primary sample	Optional weather	Primary row inclusion independent of weather
Conformal order	Prequentiality	Future target perturbation cannot change earlier radii
Conditional coverage	Hour tests	Raw and Holm-adjusted counts reconcile
Trading evaluation	Right-censoring	Incomplete D through D+6 outcomes excluded
Paper figures	Vector outputs	One nonempty PDF for each of 14 figures
Evidence identity	Frozen data	SHA-256 matches the research report

D Validation matrix and reproduction

The shortest reproduction from the committed dataset is:

```
python -m pip install -r requirements-paper.txt
python src/qa.py
python src/models.py
python src/trading.py
python src/research.py
python src/make_figures.py
python scripts/generate_paper_artifacts.py
python -m pytest -q
bash scripts/build_paper.sh
```

The first ingestion run is deliberately separate because it calls external services and may observe revised upstream values. To rebuild from source APIs, run `python src/ingest.py` before the commands above, archive the raw responses, and expect the dataset fingerprint to change.

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